

# **Prospective Multiclass Inpatient Utilization Risk Prediction Using Longitudinal MEPS Data**

## **Module 12 Capstone Final Report**

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## Executive Summary

This capstone project developed and operationalized a machine learning framework for predicting Year-2 inpatient admission risk using Year-1 features from the Medical Expenditure Panel Survey (MEPS). The final analytic design pooled five longitudinal panels, engineered utilization and condition features from event-level and conditions-level files, and evaluated models using a strict panel-based holdout design rather than a random row split. This design choice strengthened the practical value of the project because it tests whether the model generalizes to a future unseen panel rather than merely memorizing pooled historical patterns.

The final modeling cohort contained 30,659 members after removing observations with missing Year-2 inpatient targets. The outcome was modeled as a three-class prediction problem: zero admissions, one admission, and two or more admissions. Four final models were compared on the holdout panel: Gradient Boosting (GB), CatBoost, HistGradientBoosting (HGB), and Random Forest (RF). GB produced the strongest raw recall and PR-AUC for the rare 2+ admissions class, while CatBoost provided the best overall operational balance by combining strong rare-class ranking with the highest macro-F1 and top-decile lift. CatBoost was therefore selected as the final deployment model.

On the unseen holdout panel, CatBoost achieved 0.720 accuracy, 0.581 balanced accuracy, 0.409 macro-F1, 0.526 recall for the rare 2+ admissions class, and 6.41 top-decile lift. The top 10% of members ranked by CatBoost contained an observed 2+ admission rate of 8.71% versus a base rate of 1.36%, and the top 5% cohort yielded a 13.24% observed rate. The dashboard translates these results into a deployable analyst interface by combining saved model artifacts, diagnostic charts, and export-ready high-risk cohort views. The final deliverable therefore supports both technical validation and operational use.

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## Introduction

Hospital admissions are a high-cost and operationally significant component of healthcare utilization. In the Medicare and broader U.S. healthcare environment, a relatively small share of members account for a disproportionate share of inpatient use and downstream spending (Cohen & Yu, 2012). For that reason, prospective identification of members likely to require future inpatient care has value for care management, resource planning, and population health intervention. A useful operational model must do more than achieve high overall accuracy; it must identify the rare, high-risk cohort early enough to support action.

This project addresses that problem using the Medical Expenditure Panel Survey (MEPS), a nationally representative longitudinal survey that links person-level demographic, health status, utilization, and expenditure information across two years for each panel (Agency for Healthcare Research and Quality [AHRQ], 2024). MEPS is especially well suited to this capstone because it supports a panel-based Year-1 to Year-2 prediction design while remaining publicly reproducible. The present study uses Year-1 demographic, socioeconomic, self-reported health, condition, and utilization features to predict Year-2 inpatient admission burden.

The central research question is whether machine learning models trained on pooled longitudinal MEPS panels can identify future inpatient admission risk with enough fidelity to support targeted cohort review. The project was intentionally framed as a multiclass problem rather than a simple binary inpatient-use indicator so that the final output could distinguish between zero admissions, one admission, and the particularly important 2+ admissions group. This framing better reflects real decision making because resource prioritization often depends on the severity and recurrence of future utilization rather than the presence of any hospitalization alone.

The report proceeds in the style of an academic capstone paper and implementation-ready technical deliverable. It first situates the project within the literature on high-cost patient prediction, hospitalization forecasting, and interpretable healthcare machine learning. It then documents the MEPS cohort construction, feature engineering, panel-based validation design, and model comparison strategy. Finally, it presents the final results, explains the rationale for selecting CatBoost as the operational model, and describes the Streamlit dashboard and artifact-backed deployment workflow.

### **Literature Review**

A substantial healthcare analytics literature has shown that spending and utilization are highly concentrated within relatively small patient subgroups. Cohen and Yu (2012) demonstrated the concentration and persistence of healthcare expenditures over time, which helps explain why prospective risk identification remains an important analytics problem for health systems and payers. If high-utilization members can be identified earlier, resources for care coordination and intervention can be targeted more effectively.

Machine learning has become increasingly prominent in this domain because healthcare utilization is shaped by nonlinear interactions among demographic, clinical, behavioral, and historical-use factors. Bertsimas et al. (2008) showed that algorithmic models could improve healthcare cost prediction beyond simpler baselines, while Langenberger et al. (2023) found meaningful performance differences across model families in high-cost patient prediction using claims data. Zou et al. (2023) similarly reported that hybrid machine learning approaches can improve cost prediction when the predictor-outcome relationship is complex. These studies justify the present project's decision to compare multiple model families rather than assuming that a single method will dominate.

Hospitalization prediction presents additional challenges because the target outcome is both temporally structured and imbalanced. Yi et al. (2022) showed that machine learning can support planning for hospitalization-related population health risk, reinforcing the value of risk stratification for prospective intervention. Orhan and Kurutkan (2025) further argued that healthcare demand prediction benefits from jointly modeling predisposing, enabling, and need factors, a finding that maps well to the present project's inclusion of socioeconomic, self-reported health, and prior-utilization variables. In a practical sense, these studies support an approach that combines lagged utilization with broader social and clinical context.

Interpretability remains a necessary condition for operational adoption. Obermeyer and Emanuel (2016) noted that high-performing models are not enough on their own; decision-makers need intelligible explanations to translate predictions into action. This requirement is especially relevant for stakeholder-facing tools. CatBoost offers a strong fit for this requirement because it performs well on mixed tabular data and supports transparent feature importance analysis (Prokhorenkova et al., 2018). The dashboard layer further benefits from SHAP-compatible reasoning and ranked-cohort interpretation rather than black-box probability scores alone (Lundberg & Lee, 2017). Together, this literature supports the design of an interpretable, panel-validated, machine-learning workflow for inpatient risk prediction.

Recent forecasting work has also emphasized blended machine-learning and statistical approaches for healthcare expenditure and demand prediction. Wang et al. (2024) and Wang et al. (2025) both showed that hybrid forecasting designs can improve healthcare expenditure prediction, while Wu et al. (2022) demonstrated that two-stage ensemble learning can strengthen expenditure modeling in healthcare settings. These studies are relevant because the present project also prioritizes ranked risk stratification over a single-metric optimization.

Methodological work on MEPS itself also supports the present design. McClellan et al. (2023) showed that machine-learning algorithms can improve imputation within MEPS, reinforcing the broader value of learning-based methods on survey-structured healthcare data. Emerging representation-learning studies such as GraphCare and dynamic healthcare embeddings further illustrate where patient-state modeling may go next (Pengcheng et al., 2023; Jang et al., 2023), even though those architectures were beyond the scope of this capstone. Fairness remains an important constraint on operational adoption as well, and Schuch et al. (2023) provided a useful reminder that healthcare machine-learning systems should be assessed not only for predictive quality but also for equitable behavior across groups.

### **Research Methods**

The project used pooled longitudinal MEPS panels 23 through 27, corresponding to Year-1 / Year-2 mappings from 2018-2019 through 2022-2023. The raw event-enhanced analytic file contained 31,138 records. Negative survey codes indicating inapplicable, refused, not ascertained, or nonresponse values were recoded to missing. Observations lacking the Year-2 inpatient admission target were removed, yielding a final modeling cohort of 30,659 members. The final holdout design reserved Panel 27 as the unseen evaluation cohort and trained the models on Panels 23 through 26. This panel-based split was chosen specifically to reduce information leakage and to approximate a future-cohort deployment scenario.

The target variable, Y\_IPDISY2\_CLASS, was defined as a three-level categorization of Year-2 inpatient admissions: 0 admissions, 1 admission, and 2 or more admissions. The predictor matrix combined traditional MEPS variables with engineered utilization and condition features. Included domains were demographics, race and ethnicity, education, insurance, poverty category, self-reported physical and mental health, chronic condition flags, Year-1 inpatient admissions,

Year-1 prescription expenditures, total Year-1 spending, and engineered counts and costs derived from Year-1 emergency, inpatient, and prescription event files. After dropping variables with more than 50% missingness and simplifying age to a single age feature, the final model used 39 predictors.

Four final models were evaluated: Gradient Boosting, HistGradientBoosting, Random Forest, and CatBoost. Because the rare 2+ admissions class represented only 1.50% of the overall cohort and 1.36% of the holdout panel, the training workflow used class-balancing logic to avoid a majority-class model dominating the objective. Evaluation focused on metrics suitable for imbalanced multiclass prediction: accuracy, balanced accuracy, macro-F1, recall for the 2+ class, class-2 PR-AUC, and top-decile lift. The final dashboard also exposed the confusion matrix, precision-recall behavior, calibration, and holdout cohort ranking outputs to ensure that model selection remained operational rather than purely statistical.

The implementation stack used Python, pandas, NumPy, scikit-learn, CatBoost, matplotlib, and Streamlit. During finalization, the dashboard was refactored to support artifact-backed inference for the default project dataset. The model builder now saves the final CatBoost model in native format and the remaining benchmark models as compressed joblib bundles together with feature schema files, metrics, and metadata. This separation between training and serving makes the final deliverable reproducible and more realistic as an operational tool.

## **Analysis Results and Discussion**

The final analytic cohort summary is presented in Table 1, while the class distribution by panel appears in Figure 1. The cohort retained 30,659 members across Panels 23 through 27 after removing missing Year-2 targets from the 31,138-row event-enhanced file. The outcome was strongly imbalanced: 92.57% of members had no Year-2 inpatient admissions, 5.93% had one



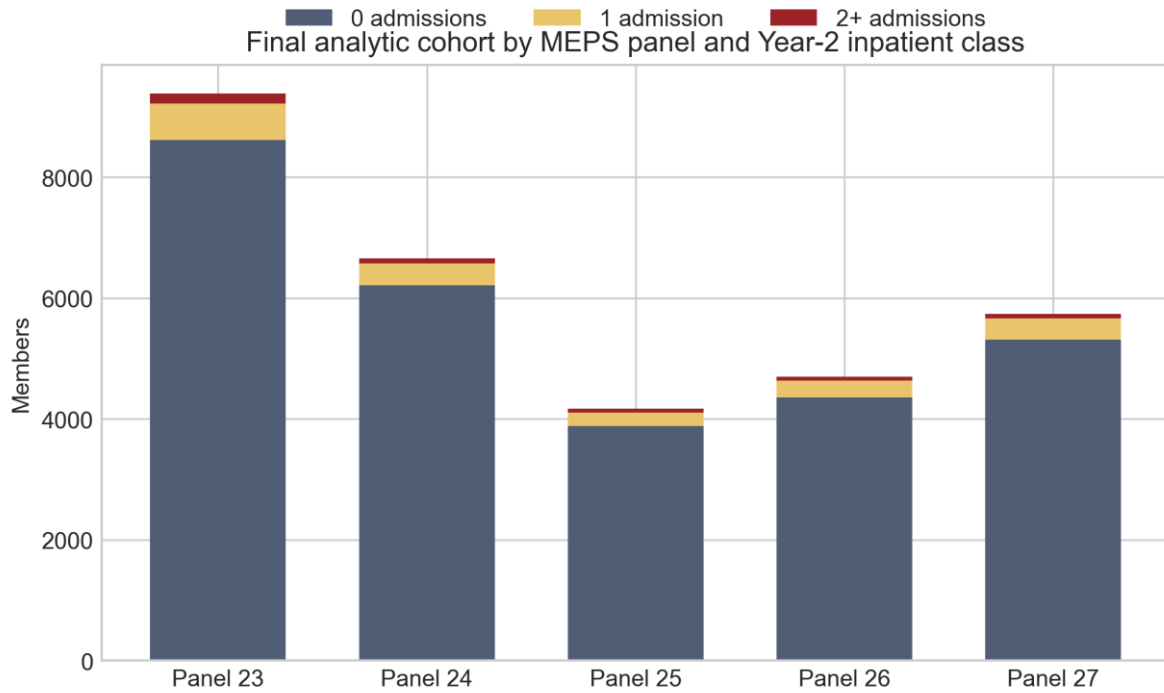
admission, and only 1.50% had two or more admissions. The holdout panel preserved this challenge, with 5,740 members and a 2+ admissions base rate of 1.36%. This imbalance is important because it makes overall accuracy a poor basis for model selection; a model can appear highly accurate while failing to capture the rare but operationally important high-risk class.

The comparative model results appear in Table 2 and Figure 2. Random Forest achieved the highest raw accuracy at 0.925, but this was a textbook example of an accuracy trap. Its balanced accuracy fell to 0.333 and its recall for the 2+ admissions class was 0.000, meaning that it effectively ignored the rare class. By contrast, Gradient Boosting produced the strongest raw rare-class capture, with the highest recall for the 2+ class (0.603), the highest balanced accuracy (0.590), and the strongest class-2 PR-AUC (0.156). HistGradientBoosting was competitive but did not lead on any key operational metric. CatBoost achieved nearly the same rare-class ranking quality as GB while producing the best macro-F1 (0.409) and the highest top-decile lift (6.410).

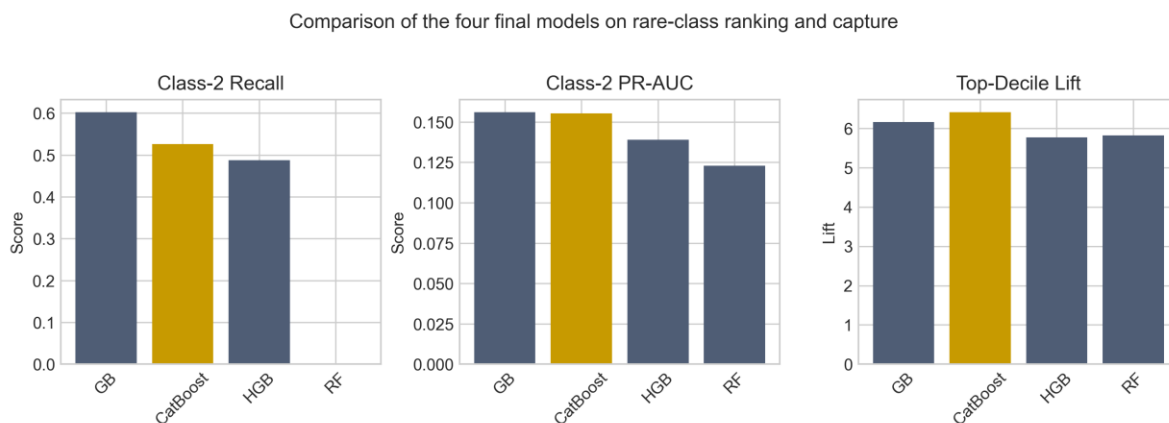
This tradeoff drove the final model selection decision. If the sole objective were to maximize class-2 recall, Gradient Boosting would be the preferred model. However, the intended use case requires a model that balances rare-class capture, multiclass stability, and ranking utility for targeted review. CatBoost was therefore selected as the final deployment model because it offered the strongest overall operational balance. This decision is defensible from both an academic and implementation perspective: CatBoost remained highly competitive on PR-AUC and recall while generating the best concentration of true high-risk cases in the top-ranked cohort.

**Table 1. Final analytic cohort summary**

| Measure                    | Value                       |
|----------------------------|-----------------------------|
| Raw event-enhanced records | 31,138                      |
| Final modeling cohort      | 30,659                      |
| Panels included            | 23, 24, 25, 26, 27          |
| Year mapping               | 2018-2019 through 2022-2023 |
| Training rows              | 24,919                      |
| Holdout rows               | 5,740                       |
| Final predictors           | 39                          |
| Overall 2+ admissions rate | 1.50%                       |
| Holdout 2+ admissions rate | 1.36%                       |

*Figure 1. Final analytic cohort by MEPS panel and Year-2 inpatient class.***Table 2. Final holdout model comparison**

| Model    | Accuracy | Balanced Accuracy | Macro-F1 | Recall (2+) | PR-AUC (2+) | Top-Decile Lift |
|----------|----------|-------------------|----------|-------------|-------------|-----------------|
| GB       | 0.689    | 0.590             | 0.391    | 0.603       | 0.156       | 6.154           |
| CatBoost | 0.720    | 0.581             | 0.409    | 0.526       | 0.155       | 6.410           |
| HGB      | 0.724    | 0.549             | 0.396    | 0.487       | 0.139       | 5.769           |
| RF       | 0.925    | 0.333             | 0.320    | 0.000       | 0.123       | 5.820           |



*Figure 2. Comparison of the four final models on rare-class ranking and capture.*

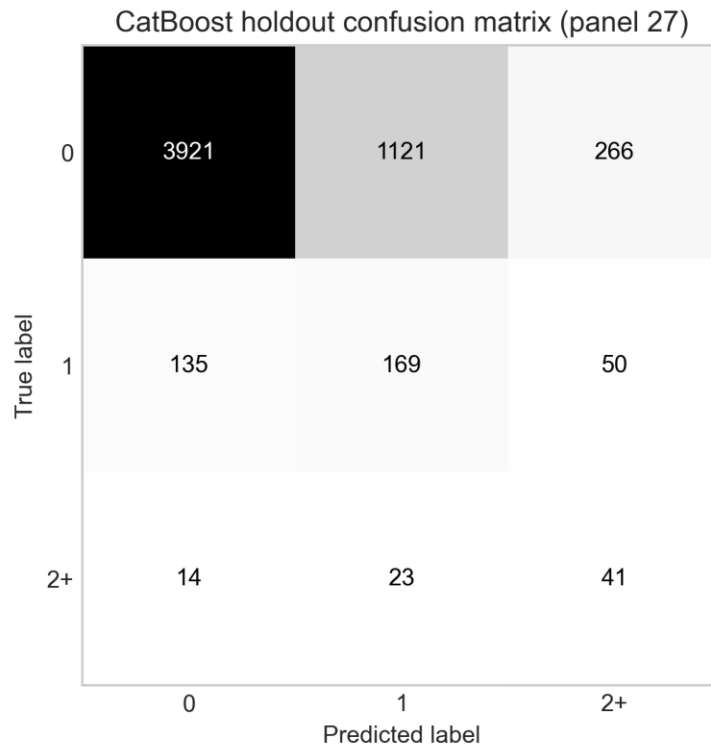
Figure 3 shows the CatBoost confusion matrix on the unseen holdout panel. The model correctly classified 41 of the 78 members in the 2+ admissions class through strict class assignment, while still maintaining reasonable separation for the majority and one-admission classes. This is not perfect rare-class recall, but it is materially better than Random Forest and sufficiently strong when combined with ranked-cohort use. In practice, the more important insight is that the probability scores remain useful even when exact class assignment is conservative.

That ranked-cohort behavior appears clearly in Figure 4. The top-decile holdout cohort ranked by CatBoost probability had an observed 2+ admissions rate of 8.71%, which is 6.41 times the panel base rate. The top 5% cohort performed even more strongly, with 38 of the holdout panel's 78 true 2+ admission cases appearing among only 287 members. That corresponds to an observed rate of 13.24% and a lift of 9.74. In operational terms, the dashboard allows analysts to concentrate review on a much smaller cohort while preserving a large share of the rare high-risk cases.

The calibration curve in Figure 4 also surfaces an important caution. CatBoost's class-2 probabilities are directionally informative but over-confident at higher score levels. That means the probabilities are more defensible as ranking scores than as literal probability estimates for

resource budgeting. This is a manageable issue, not a fatal one. In many operational settings, ranking utility matters more than perfect probability calibration because the downstream decision is to prioritize members for review or intervention rather than to quote an exact risk percentage.

Figure 5 presents the leading CatBoost predictors. The strongest signals were age, multiple self-reported physical health categories, prior total expenditure, sex, prescription-use intensity, condition burden, and insurance coverage. These results align with the broader literature on healthcare demand prediction, which consistently finds that age, prior utilization, chronic disease burden, and self-reported health status contribute meaningfully to future use (Bertsimas et al., 2008; Langenberger et al., 2023; Orhan & Kurutkan, 2025). The final feature set also demonstrates the value of combining survey-reported health and socioeconomic variables with engineered utilization summaries from event and condition files.



*Figure 3. CatBoost holdout confusion matrix on panel 27.*

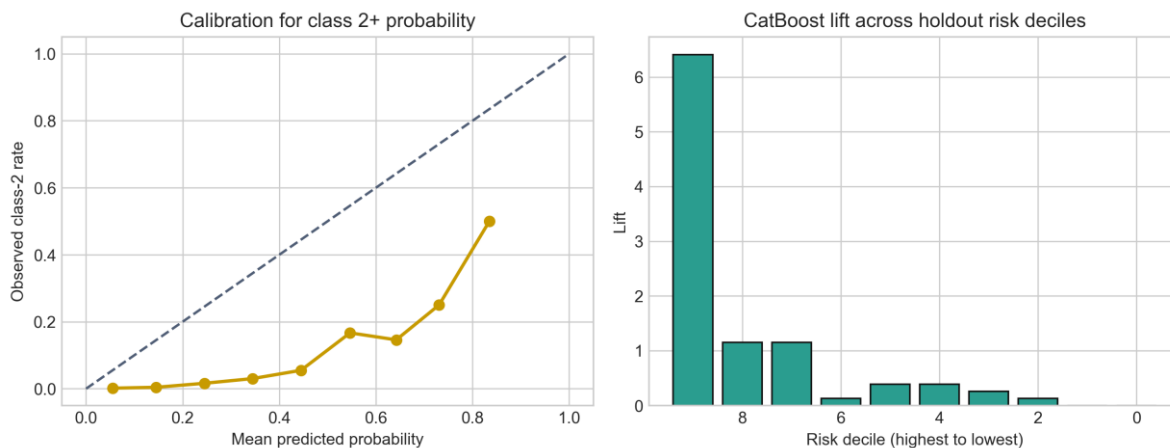


Figure 4. Calibration and lift diagnostics for CatBoost on the holdout panel.

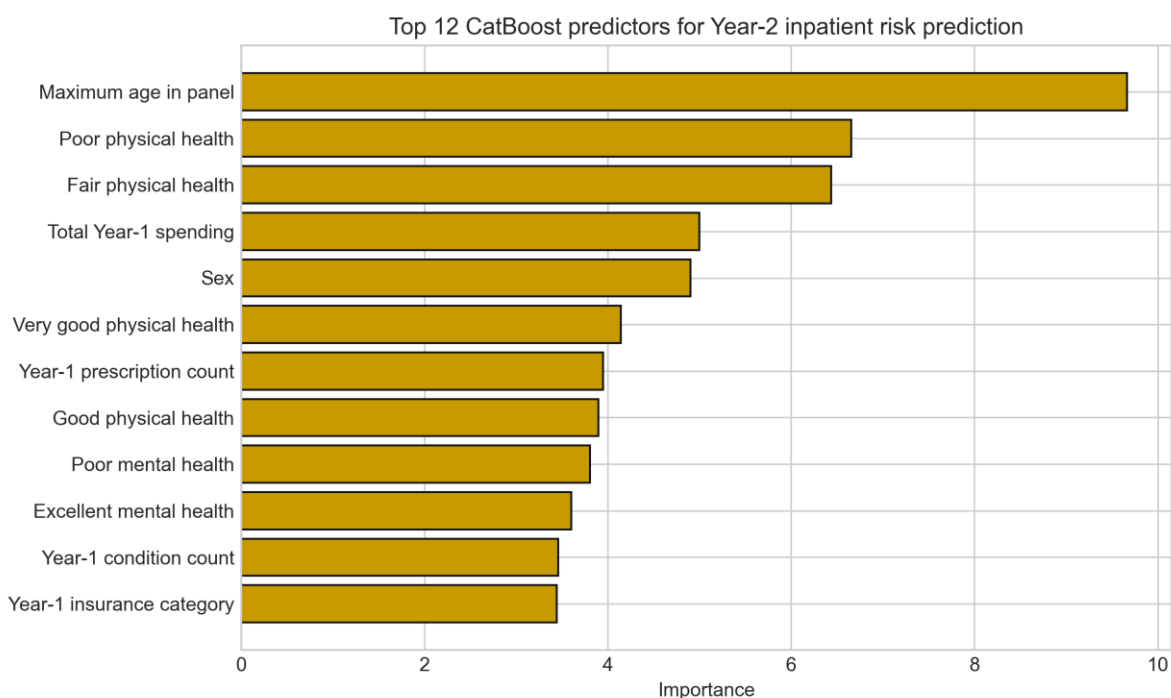


Figure 5. Top 12 CatBoost predictors for Year-2 inpatient risk.

## Dashboard Implementation

The Streamlit dashboard was designed as an analyst-facing decision-support tool rather than a model-development notebook. The interface exposes four classes of functionality: configuration controls, model comparison, diagnostic interpretation, and high-risk cohort export.

Sidebar controls allow the user to choose the model set, specify the ranked cohort size, and optionally upload a new analysis-ready parquet file. The main page surfaces cohort size, holdout size, feature count, and panel-based validation configuration to make the evaluation design explicit.

A key finalization step was the introduction of artifact-backed inference. For the default project dataset, the dashboard now loads the saved CatBoost, GB, HGB, and RF artifacts directly from the artifacts/models directory rather than retraining all models on every run. This change materially improves reproducibility and aligns the dashboard with realistic deployment practice. It also preserves a flexible extension path: when a user uploads a new parquet file, the dashboard falls back to runtime training so the same interface can still support experimentation or later-year data refreshes.

The middle layer of the dashboard emphasizes model governance and interpretation. Users can review the full holdout comparison table, inspect the confusion matrix and precision-recall behavior for a selected model, assess calibration and lift, and switch directly to ranked or consensus cohort views. Because the tool is intended for high-risk cohort review rather than fully automated action, the design explicitly supports downloadable CSV outputs and ranked-case review rather than one-click intervention logic.

From an implementation perspective, the dashboard's most important contribution is that it bridges the gap between model evaluation and analyst workflow. The saved artifacts, metadata, and feature schema files ensure that the delivered model is not just a notebook result but a portable inference bundle. The dashboard therefore functions as both a reporting surface and a lightweight deployment layer for continued cohort review.

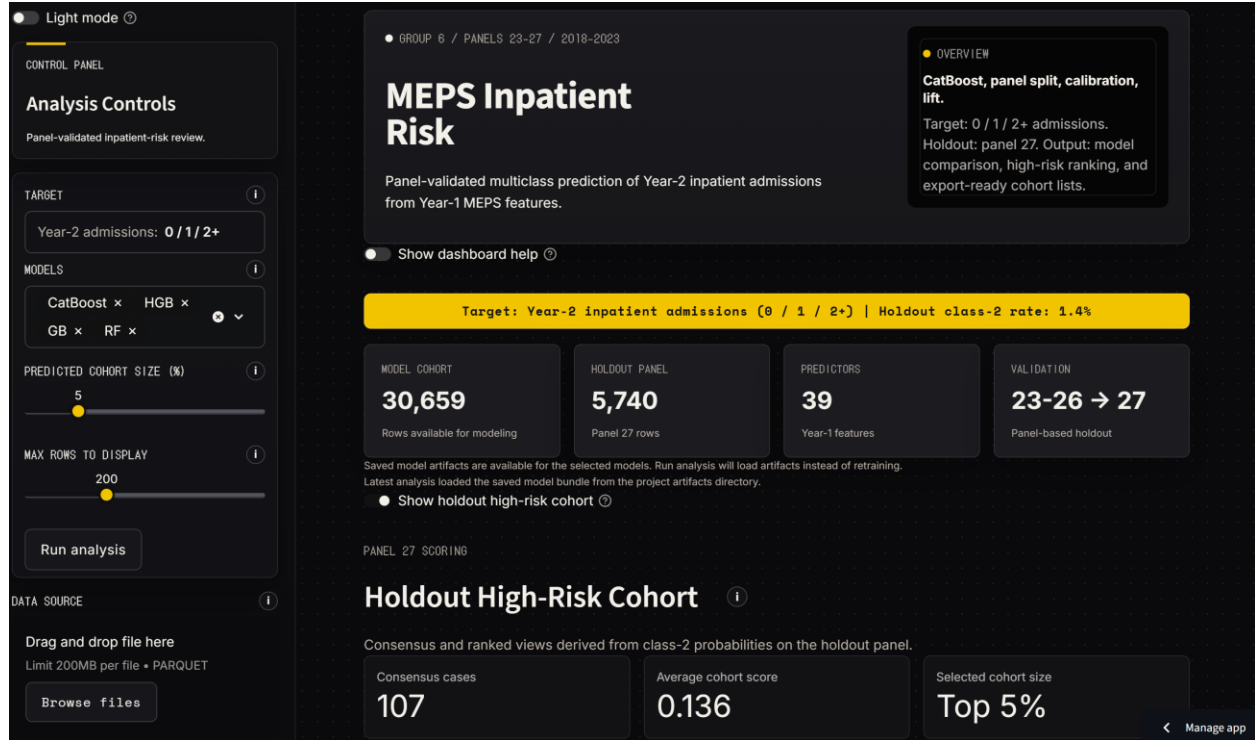


Figure 6. Dashboard landing view and analysis controls

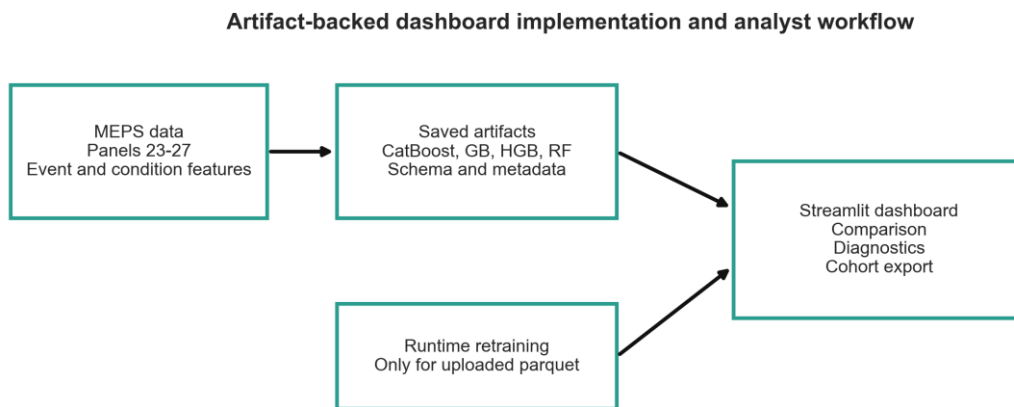


Figure 7. Artifact-backed dashboard implementation and analyst workflow.

**Table 3. Dashboard components and analyst tasks**

| Dashboard Component    | Primary Purpose                                                              | Analyst Action                                                        |
|------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Sidebar controls       | Select models, cohort size, and optional uploaded parquet                    | Configure the analysis run.                                           |
| KPI cards              | Summarize cohort size, holdout panel, predictor count, and validation design | Confirm the evaluation context before interpreting results.           |
| Model comparison table | Compare multiclass and rare-class metrics across final models                | Choose the final model and justify deployment.                        |
| Diagnostics panel      | Show confusion, PR behavior, calibration, and lift                           | Assess tradeoffs between ranking quality and classification behavior. |
| Holdout cohort export  | Provide ranked and consensus case lists                                      | Download high-risk cohorts for follow-up review.                      |
| Artifact status note   | Indicate whether the dashboard loaded saved models or retrained              | Verify reproducibility and serving mode.                              |

## Conclusion

This capstone project demonstrates that longitudinal MEPS data can support a defensible approach to inpatient risk prediction when the design emphasizes leakage control, panel-based validation, and operational ranking metrics. The final results showed that raw accuracy is an inadequate selection criterion in this setting, because Random Forest achieved the highest accuracy while failing to identify the rare high-risk class. CatBoost provided the strongest overall deployment balance by combining strong class-2 ranking with the highest macro-F1 and top-decile lift. The resulting dashboard turns those model outputs into an interpretable workflow for holdout cohort review and export. The project therefore satisfies both the analytical and implementation goals of the capstone.



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## Appendix A. Predictor Inventory

Table 4 groups the final 39 predictors into the main domains used in the final model. This appendix supports the methods section without overloading the main narrative.

**Table 4. Final predictor domains and representative features**

| Predictor Domain               | Representative Features                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Demographics                   | Maximum age in panel; sex                                                                                                               |
| Race / ethnicity               | Race and ethnicity indicators drawn from the longitudinal person file                                                                   |
| Socioeconomic context          | Years of education; poverty category; Year-1 insurance category                                                                         |
| Self-reported health           | Physical health indicators from excellent to poor; mental health indicators from excellent to poor                                      |
| Condition burden               | Diabetes, arthritis, asthma, injury flags, chronic-condition count, total condition count, ICD-10 diversity                             |
| Prior utilization and spending | Total Year-1 spending; prescription spending; prior inpatient use; Year-1 prescription, inpatient, and emergency event counts and costs |

## Appendix B. Saved Model Artifact Inventory

Table 5 inventories the persisted model, preprocessing, and metadata files that support the deployed dashboard.

**Table 5. Saved model artifacts used by the dashboard**

| Artifact File            | Purpose                                                                      |
|--------------------------|------------------------------------------------------------------------------|
| catboost_model.cbm       | Native final CatBoost model.                                                 |
| catboost_bundle.joblib   | CatBoost preprocessing metadata and feature order.                           |
| gb_bundle.joblib         | Gradient Boosting model bundle.                                              |
| hgb_bundle.joblib        | HistGradientBoosting model bundle.                                           |
| rf_bundle.joblib         | Random Forest model bundle.                                                  |
| feature_columns.json     | Exact feature schema used in training.                                       |
| categorical_columns.json | Categorical feature list used in preprocessing.                              |
| model_metadata.json      | Training panels, holdout panel, row counts, versions, and target definition. |
| model_metrics.csv        | Saved holdout comparison metrics for the final model set.                    |

## Appendix C. Dashboard Screens

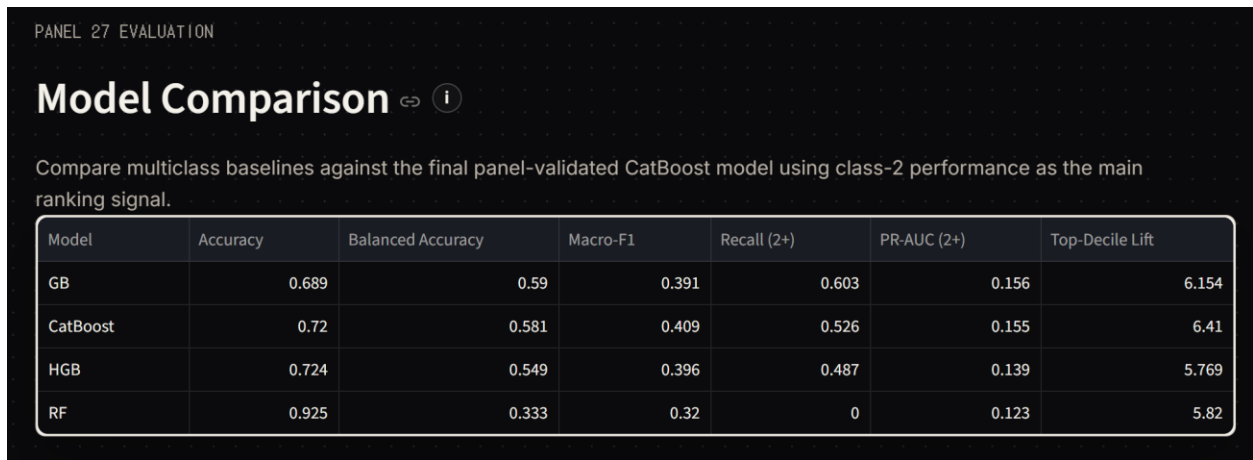


Figure C1. Final model comparison dashboard.

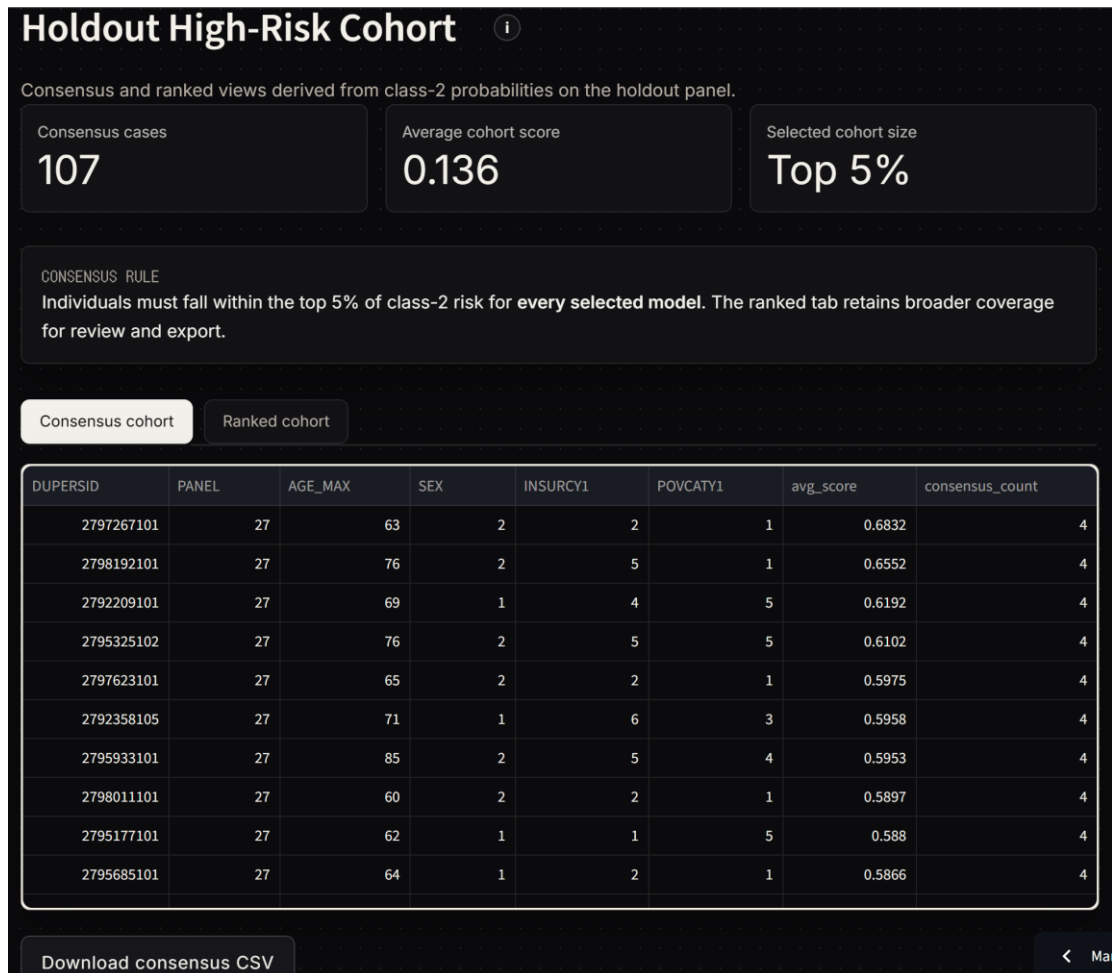


Figure C2. Holdout cohort export view.